Department of Civil and Environmental Engineering Autumn 2026
CET 513 Transportation Networks and Optimization **Dr. Jeff Ban**

Course Project

You are supposed to form a ***four-person team*** to work on a project that has an optimization / network flow component. CEE or other types of engineering applications are preferred, but not necessary.

Selecting specific topics that are relevant to your own research area is highly encouraged.

Milestones & Deliverables:

1. Oct. 26: Check point #1: form your team and topic (email or one pager to Instructor).
2. Nov. 09: Check point #2: Topic finalized and start the work.
3. Nov. 25: Check point #3: good progress of the project
4. Dec. 09: Student presentations. 15 minutes presentation and 3 minutes Q&A
5. Dec. 13: Final Report Due at 12 pm

Your report should minimally include the followings (12 – 20 pages):

1. Title of the project
2. Team members and a brief description of the responsibility of each member
3. Problem statements and motivation
4. Literature review
5. Research methodology
6. Major results, analysis, findings, and recommendations
7. Conclusions and suggestions for future research directions
8. Reference List
9. Author (team member) contribution statement (a brief paragraph about the role and contributions of each team member to the project)
10. AI use statement: how AI tools are used in the project and in the report preparation

Your final report should be prepared as a complete research paper, following the manuscript format used by Transportation Research Record: The Journal of Transportation Research Board. See https://trb.secure-platform.com/a/page/TRBPaperReview, especially the Sample Paper. You can ignore the TRB restrictions on # of words, etc.

Submissions: In addition to your project report, please also upload your presentation slides, data you used if you can share (or a link if the size of the data is too large), and the computer codes you developed to implement your model.